

Composing Complex Numerals in Mandarin

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1. Introduction

In this talk, I provide a semantics that is syntactically informed for Mandarin numerals.

The focus of the analysis is not on the meaning of numerals per se, but rather how complex numerals relate to simplex numerals, and how complex numerals are interpreted in numeral-classifier constructions.

- First, I show that numeral items in Mandarin fall into two syntactic classes, Num^o and N^o (§ 2).
- Second, I show how complex numerals can be built in syntax (§ 3).
- Last, I provide a semantics of relating complex numerals to its building blocks, and their interpretation under numeral-classifier constructions (§ 4).

2. The Syntax of Mandarin Numerals

It is usual for the system of numeral items in a language to show divided behaviours (KAYNE 2019). This is indeed the case in Mandarin.

Table 1 shows the list of numeral items in Mandarin.

- I call those that correspond to numbers in $\llbracket 1, 10 \rrbracket$ *small numerals*;
- and bases $\{10^k \mid k \in \{2, 3, 4, 8, 12\}\}$ *ten's powers* (excluding ten!).

Small Numerals										Ten's Powers				
<i>yī</i>	<i>èr</i>	<i>sān</i>	<i>sì</i>	<i>wǔ</i>	<i>liù</i>	<i>qī</i>	<i>bā</i>	<i>jiǔ</i>	<i>shí</i>	<i>bǎi</i>	<i>qiān</i>	<i>wàn</i>	<i>yì</i>	<i>zhào</i>
1	2	3	4	5	6	7	8	9	10	10 ²	10 ³	10 ⁴	10 ⁸	10 ¹²

Table 1: Numeral Items in Mandarin

Small numerals and ten's powers display very distinct syntactic properties. The distinction holds across both arithmetical and counting uses.

- Ten's powers but not small numerals always need the support of of a multiplier to express their base values.
 - Compare (1) with (2)
- Ten's powers but not small numerals, when used as multiplicands in multiplicative complex numerals, can trigger suppletion on 'one' and 'two'.¹
 - Compare (3) with (4) (for space reasons I only include examples involving suppletive 'two')
 - I take such suppletions to be reflexion of Num^o's agreement with Cl^o.

(1) Ten's powers always need the support of a multiplier *yī* 'one' to express their base values.

- a. **(yì) {bǎi/ qiān} kē píngguǒ*
 one hundred thousand CL apple
 'one {hundred/ thousand} apples'
- b. **(Yì) {bǎi/ qiān} shì dà shùzì.*
 one hundred thousand COP big number
 'One {hundred/ thousand} is a big number.'

(2) Small numerals do not need multiplier *yī* 'one' to express their base values.

- a. *{sān/ shí} kē píngguǒ*
 three ten CL apple
 '{three/ ten} apples'
- b. *{Sān/ Shí} shì xiǎo shùzì.*
 three ten COP small number
 '{Three/ Ten} is a small number.'

(3) Ten's powers can trigger suppletion on multiplier 'two'

- a. *{(?èr/ {^{OK}liǎng}) {bǎi/ qiān} kē píngguǒ.*
 two two.SUPP hundred thousand CL apple
 'two {hundred/ thousand} apples'
- b. *{(?Èr/ {^{OK}Liǎng}) {bǎi/ qiān} shì dà shùzì.*
 two two.SUPP hundred thousand COP big number
 'Two {hundred/ thousand} is a big number.'

¹See Wu (2024) for why 'one' also has a suppletive form.

(4) Small numerals do not trigger suppletion on multiplier ‘two’.

- a. {èr/ *liǎng} shí kē píngguǒ
 [two] two.SUPP ten CL apple
 ‘twenty apples’
- b. {Èr/ *Liǎng} shí shì xiǎo shùzì.
 [two] two.SUPP ten COP small number
 ‘Twenty is a small number.’

Simplex numerals (which can only be small numerals since ten’s powers always need the support of a multiplier), or complex numerals (which are built from small numerals plus ten’s powers) never have predicative use, which speaks against the adjectival analysis.

❑ Regardless of whether there is a copula, numerals never have predicative use in Mandarin (5).

❑ The only way to remedy (5) is to use (existential) ‘be’, and add a classifier after the numeral (6).

(5) No predicative use of numerals.

*Xuéshēng (shì) {sān/ shí-èr/ liǎng-bǎi}.
 student COP three ten-two(=twelve) two-hundred
 Intended: ‘There are {three/ twenty/ two hundred} students.’

(6) Remedy to the ungrammaticality of (5): to use *yǒu* and add a classifier.²

Xuéshēng yǒu {sān/ shí-èr/ liǎng-bǎi} ×(míng).
 student be three ten-two(=twelve) two-hundred CL
 ‘There are {three/ twenty/ two hundred} students.’

❑ The requirement of a classifier in (6) suggests that the sentence involves movement of NP from [Num Cl NP].

❑ which in turn suggests that all numerals under counting uses come hand-in-hand with classifier-noun’s.

❑ We will see in § 3 that all numerals, no matter used arithmetically or in counting, co-occur with classifier-noun’s.

Based on these facts I propose that

❑ Small numerals are exclusively functional elements, merging as Num°.

❑ Ten’s powers are exclusively N°, which themselves combine with Num° (and Cl°).

²The marker × suggests inappropriateness. Without a classifier, (6) may not be too bad as “ungrammatical”, but makes you sound like you are from 10th century (especially with simplex numerals).

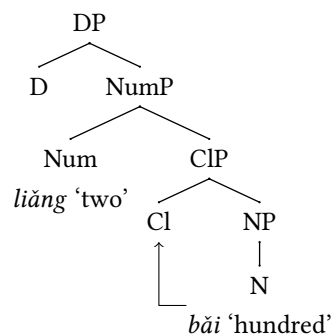
3. Building Complex Numerals in Syntax

Multiplicative complex numerals fall into three classes.

❑ The easiest case is that the multiplicand is a ten’s power and the multiplier is a small numeral. Then they combine similar to how ‘two’ and ‘apple’ combine

- See (7), where I adopt the Mandarin DP structure by Tang (1990) a.o.. I also assume that ten’s powers need to move to Cl° (SIMPSON, SOH & NOMOTO 2011), for reasons I will go through today.
- In some sense, ‘two hundred’ is structurally no different from ‘two apples.’

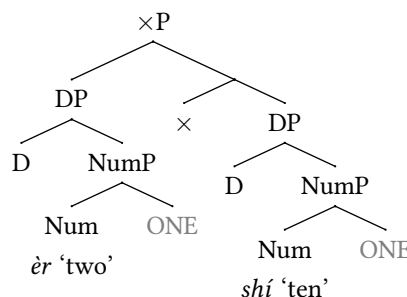
(7) Multiplicative complex numerals, base=ten’s power, multiplier=small numeral.
liǎng bǎi ‘two hundred’



❑ If the multiplicand numeral is a small numeral (which can only be ten), and the multiplier also a small numeral, then they are respectively full DP’s.

- See (8).
- Each numeral takes a silent NP, denoted as ONE, as its complement, much in the spirit of Moltmann (2013).³

(8) Multiplicative complex numerals, base=small numeral, multiplier=small numeral.
èr shí ‘two ten (=twenty)’



³See also Kayne (2005a), Collins (2007), Moltmann (2022) i.a. for the notion of light noun.

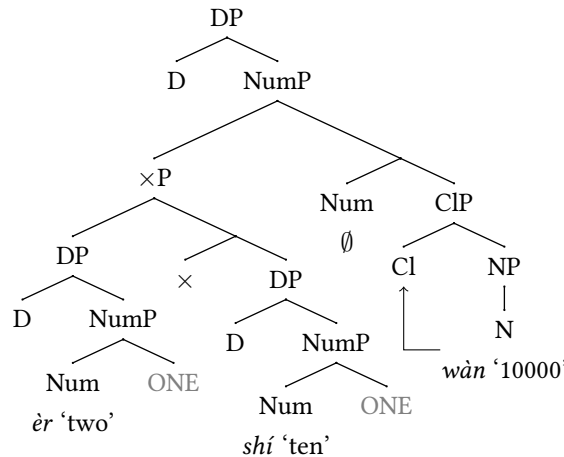
- ❑ If the multiplicand is a ten's power and the multiplier itself is a complex numeral, then the multiplier sits in Spec,NumP, the head of which is empty.

➤ See (9).

➤ While the multiplier in this example is a $\times P$, it can as well be $+P$ or DP.

- (9) Multiplicative complex numerals, multiplicand=small numeral, multiplier=complex numeral.

èr shí wàn '20*10000'

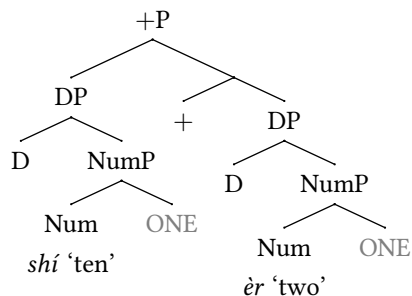


For additive complex numerals, the added two subparts are always full-fledged DP's themselves, with the relating head being $+^{\circ}$.

- ❑ See (10).

- ❑ It deserves to be thought whether $\times^{\circ}$ should be collapsed $+^{\circ}$ into a single head in syntax, with their different semantics's attributed to encyclopædic knowledge.

- (10) *shí èr* 'ten two (=twelve)'



Further complex numerals can be built through recursive merger of $\times^{\circ}$ and $+^{\circ}$ and Num $_{\emptyset}$.⁴

⁴Actually only $+^{\circ}$ and Num $_{\emptyset}$, since multiplying two already-complex numerals in Mandarin is not felicitous.

Under this analysis, the set of numeral items in Mandarin is split into the Num $^{\circ}$ -kind (small numerals) and the N $^{\circ}$ -kind (ten's powers), each of them has consistent syntactic properties, in the spirit of antihomophony (EMBICK 2003, KAYNE 2022).

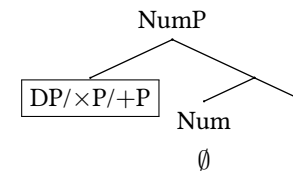
Now I can relate the arithmetic use and the counting use of numerals.

- ❑ For complex numerals, the counting use is simply its counting use merged in Spec,NumP, with the head Num being empty.

- ❑ For simplex numerals, in counting they are the Num $^{\circ}$ heads. Arithmetically they are DP which involve [Num ONE], which is very desirable since the structure [Num ONE] is now not motivated only for being embedded under complex numerals such as (8, 10), but it has its own independent use.

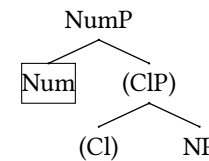
- (11) Complex numerals.

- a. Arithmetical use: DP/ $\times P$ / $+P$, such as shown in (7, 8, 9, 10)
b. Counting use: DP/ $\times P$ / $+P$ (in Spec,NumP)



- (12) Simplex numerals

- a. Arithmetical use: DP i.e. [DP[NumP[Num ONE]]]
b. Counting use: Num $^{\circ}$ taking regular classifier-noun



While the constituency status of complex numerals in counting in Mandarin has been assumed as correct a priori in my analysis, we have now arrived at this from a different angle.

- ❑ See He (2015) for independent evidence for constituency of complex numerals.

Under my analysis, complex numerals are phrases while simplex numerals are heads, and therefore it is syntactically expected for them to be merged differently, i.e. (11b) versus (12b), in a numeral-classifier construction.

- ❑ If complex numerals are constituents and in counting merge in a different position from simplex numerals, then we expect to see some syntactic differences between [Comp-Num Cl N] and [Simp-Num Cl N].

- This is exactly what we see in the silent YEAR issue (see KAYNE 2005b) in Mandarin (13), which extends also to Korean (14).
- Under an Ionin & Matushansky-style analysis, where e.g. the structure of “18 apples” contains “8 apples”, the patterns in (13, 14) would be unexpected.

(13) Mandarin: Simplex numerals not compatible with silent YEAR

- a. *Zhāngsān jīnnián bā.
 NAME this.year eight
 Intended: ‘Zhāngsān is eight years old this year.’
- b. Zhāngsān jīnnián shí-bā.
 NAME this.year ten-eight(=eighteen)
 ‘Zhāngsān is eighteen years old this year.’

(14) Korean: Simplex numerals not compatible with silent YEAR⁵

- a. *Chelswu-nun ohay {twul/ yeltelp} -i-ta.
 NAME-TOP this.year two eight -COP-DECL
 Intended: ‘Chelswu is {two/ eight} years old this year.’
- b. Chelswu-nun ohay {yel-yeltelp/ swuyn} -i-ta.
 NAME-TOP this.year ten-eight(=eighteen) fifty -COP-DECL
 ‘Chelswu is {eighteen/ fifty} years old this year.’

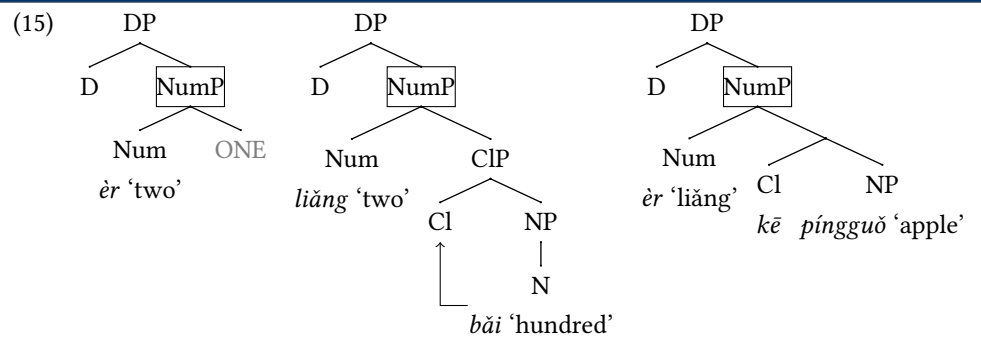
4. The Semantic Side

4.1. Part 1

I argue that no matter it is ten’s powers, or silent ONE that small numerals take as complements, they are all semantically similar to nouns. Under this view, sentences like “Four is a prime number” and “Two plus two is five” do not involve reference to mathematical numbers, but simply to entities, for Moltmann (2013) a.o. but pace Zweig (2005), Ionin & Matushansky (2018) a.o..

- The boxed nodes in (15) have parallel semantics.

⁵Tens multiples from 20 to 90 in Korean are complex (with identifiable polymorphic etymologies).



I argue that the set of individuals involve abstract entities that are for arithmetical ends.

- Ten’s powers and ONE are predicates for these abstract arithmetical entities, seen (16a) and (16b).
- There is a tiny difference. The denotation of ONE is the set of already-atomised one’s, but for ten’s powers, their denotations are yet to be atomised by Cl°.

- (16) a. $\llbracket \text{ONE} \rrbracket = \lambda x. \text{ONE}(x) \wedge \text{ATOM}(x).$
 b. $\llbracket \text{bǎi} \rrbracket = \lambda x. \text{HUNDRED}(x).$

While this talk is not about the semantics of numerals per se, I still have to have one. The semantics of èr ‘two’, for example, has the value in (17). This semantics is similar to Chierchia (2004) in Ionin & Matushansky (2018 p. 77).

- (17) $\llbracket \text{èr} \rrbracket = \lambda P. \lambda x. \exists X [x = \oplus X \wedge \#X = 2 \wedge \forall x' \in X [P(x') \wedge \text{ATOM}(x)]].$

The semantics of D° for arithmetically used numerals, as in the first two trees in (15), can simply be a choice function. Here I give an explicit algorithm.

- Suppose that abstract entities as one’s and hundred’s are all identifiable by subscripted indices.
- E.g. o_1 and o_{13} are two different one’s. h_4 and h_6 are two different hundred’s.
- I denote this special determiner, which takes sets of mereological sums of abstract arithmetical entities, as D_n .

- (18) $\llbracket D_n \rrbracket = \lambda P. \text{argmin}_{h \in P} \sum_{\substack{h' \sqsubseteq h \\ \text{ATOM}(h')}} \text{idx}(h'),$

where idx is the function that retrieves the index that a numeric unit carries.

4.2. Part 2

There are two tasks left.

- What is the compositional semantics for +P and ×P complex numerals?
- How should complex numerals be interpreted under numeral-classifier constructions? (This would also answer how complex numerals as in (9) could be interpreted.)

I propose that what $\times^\circ$ and $+\circ$ do is to combine two individuals into a new one, which I represent as vectorisation.

□ $\times^\circ$ and $+\circ$ must do combination differently such that addition and multiplication can be assessed differently.

□ I represent this difference by using two ways of vectorisation: $\times^\circ$ vectorises vertically and $+\circ$ horizontally.

□ I let S denote the set of abstract arithmetic entities, which is the domain from which $\times^\circ$ and $+\circ$ take their arguments.

$$(19) \quad \begin{aligned} \text{a. } \llbracket \times^\circ \rrbracket &= \lambda s_1. \lambda s_2. \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}. \\ \text{b. } \llbracket +^\circ \rrbracket &= \lambda s_1. \lambda s_2. \begin{bmatrix} s_1 & s_2 \end{bmatrix}. \end{aligned}$$

Definition 1. Let S_0 be the set of mereological sums of one's, hundred's, thousand's, {ten thousand}'s, {ten million}'s and trillion's. Define

$$S_i = S_{i-1} \cup \left\{ \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} a & b \end{bmatrix} : a, b \in S_{i-1} \right\}.$$

Then define

$$S = \lim_{i \rightarrow +\infty} S_i.$$

Clarification: Using vectors does not mean that they are not entities. It is a matter of representation.

□ I can as well exploitatively invent operation symbols like $\boxplus$ or $\boxtimes$, or use a collapse function (similar to $\uparrow$ in LANDMAN 1989). But this needs to be handled seriously for logical reasons.

□ I adopt a vector representation to retain transparency of the internal relationship between sub-parts.

The last thing to do is to figure out how to interpret numeral-classifier constructions involving complex numerals, e.g. the structure in (11b).

□ We need a semantics for the head Num° that relates a complex numeral (which has an entity interpretation) and a CIP (which is a set of atoms).

□ The trick is to come up with a function that can read the elements in S and returns the corresponding number.

□ Essentially, the denotation of Num° can be built from the denotation of a small numeral.

□ Compare (20) and (21). What Num° does is to take one more argument (s) than ér 'three', and replace 2 with $f(s)$.

$$(20) \quad \llbracket \text{ér} \rrbracket = \lambda P. \lambda x. \exists X [x = \oplus X \wedge \#X = 2 \wedge \forall x' \in X [P(x') \wedge \text{ATOM}(x)]].$$

$$(21) \quad \llbracket \text{Num}_\emptyset \rrbracket = \lambda P. \lambda s. \lambda x. \exists X [x = \oplus X \wedge \#X = f(s) \wedge \forall x' \in X [P(x') \wedge \text{ATOM}(x)]].$$

Definition 2. Define $f : S \rightarrow \mathbb{N}^*$, where

$$f(s) = \begin{cases} f(s_1)f(s_2) & \exists s_1, s_2 \in S \text{ such that } \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = s \\ f(s_1) + f(s_2) & \exists s_1, s_2 \in S \text{ such that } \begin{bmatrix} s_1 & s_2 \end{bmatrix} = s \\ n & \exists X \subseteq \text{ONE} \text{ such that } s = \oplus X \wedge \#X = n \wedge \forall x \in X [\text{ATOM}(x)] \\ 100n & \exists X \subseteq \text{HUNDRED} \text{ such that } s = \oplus X \wedge \#X = n \wedge \forall x \in X [\text{ATOM}(x)] \\ \vdots & \vdots \end{cases}.$$

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